

Intrinsic Timing and Position Reconstruction with Adjacent SiPMs in a 1.4 m EJ-204 Scintillator Bar

Fourth- and Twentieth-Detected-Hit Timing,
Local Position Reconstruction and EndTop Context

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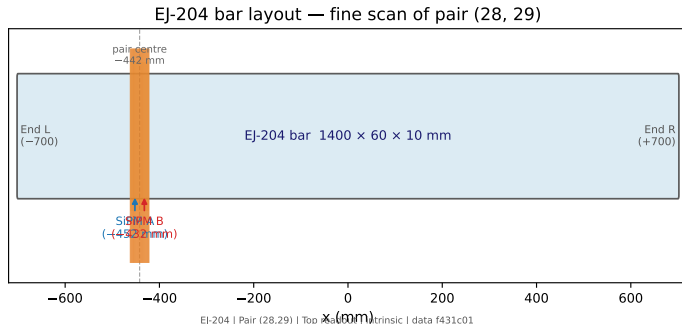
EXEC_12T-B 12 June 2026 data: f431c01

Simulation prediction — EJ-204 — Top readout — intrinsic (jitter = 0)

Scientific questions

- ① What is the intrinsic timing precision with the 4th-hit (4PE) vs. the 20th-hit (20PE)?
- ② How do efficiency, timing width, and position resolution compare for $k=4$ and $k=20$?
- ③ Does $k=20$ genuinely improve the timing or does it merely select brighter events?
- ④ What is the position sensitivity (slope, v_{eff}) of each estimator?
- ⑤ How do these results connect to the global EndTop reconstruction from EXEC_12?
- ⑥ How do our order-statistic estimators compare to the waveform-based 20% CFD method of Lv et al. (2026)?
- ⑦ What instrumental effects are still absent from this simulation?

Detector geometry and fine scan



Bar: EJ-204, $1400 \times 60 \times 10$ mm

Top SiPM array: 70 channels

Pair (28, 29):

SiPM A at $x = -452$ mm

SiPM B at $x = -432$ mm

Fine scan:

41 positions, -462 to -422 mm

1 mm steps, 3000 events each

EJ-204: $\tau_d = 2.1$ ns, $\lambda_{\text{peak}} = 408$ nm

Table: Available simulation data sets.

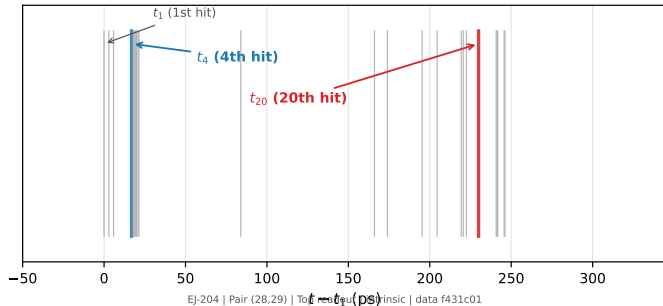
Dataset	Positions	Events/pos	Material	Readout	Jitter	Path
Fine scan (28/29)	41	3000	EJ-204	Top	0	pairsan_2026-06-11
Global EndTop	–	2000	EJ-204	End+Top	0	exec07_endtop_2000
Window-dip	4	2000	EJ-204	Top	0	exec08b_window_dip

Efficiency note

All 41 positions have 100% efficiency for both $k = 4$ and $k = 20$: $\text{NPE} \gg 20$ everywhere in the fine scan. No efficiency plot is shown because it contains no information.

From photon hits to event observables

Detected-hit sequence for one event — order statistics



For each event and each SiPM channel, sort detected hits by arrival time. Define:

$t_A^{(k)}$ = k -th smallest arrival time in channel A

Key observables:

$$\Delta t^{(k)} = t_A^{(k)} - t_B^{(k)}$$

$$t_+^{(k)} = \frac{t_A^{(k)} + t_B^{(k)}}{2}$$

- $\Delta t^{(k)}$ encodes position
- $t_+^{(k)}$ encodes event time

Timing estimators — order statistics

Order-statistic definition

For a single channel and single event, sort the detected hits:

$$t^{(1)} \leq t^{(2)} \leq \dots \leq t^{(k)} \leq \dots$$

$k=4$: $t^{(4)}$ is the 4th smallest arrival time.

$k=20$: $t^{(20)}$ is the 20th smallest arrival time.

Differential observable (position)

$$\Delta t^{(k)} = t_A^{(k)} - t_B^{(k)}$$

Calibrated as: $\mu_{\Delta t}(x) = m_k x + b_k$

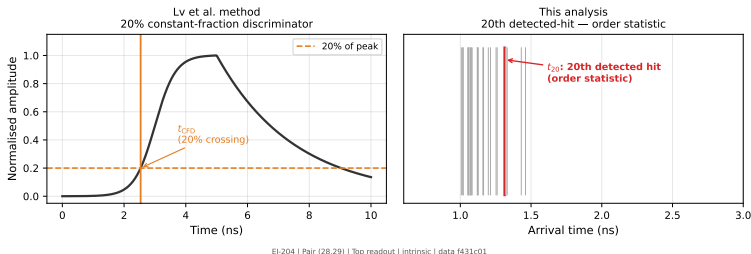
Common observable (event time)

$$t_+^{(k)} = \frac{t_A^{(k)} + t_B^{(k)}}{2}$$

Sensitive to absolute arrival time of the particle.

The 20th hit is NOT the 20% CFD

$t_{20} \neq 20\% \text{ CFD}$ — conceptually different estimators



Lv et al. (2026):

- Reconstruct full SiPM waveform
- Define t_{CFD} at 20% of peak amplitude
- Average 64 channels, weighted by NPE

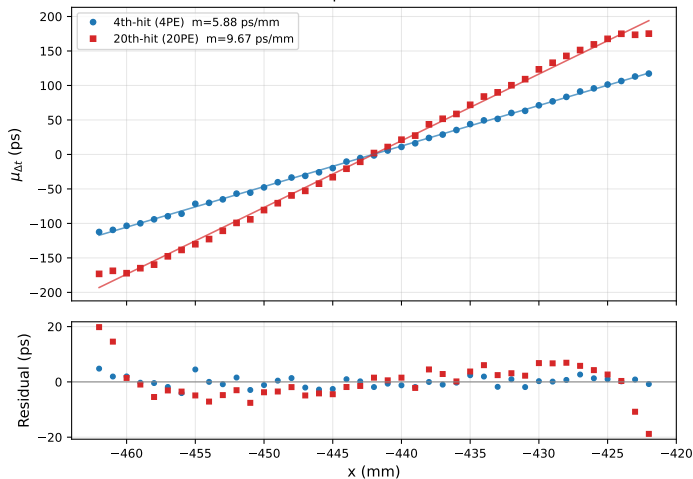
This analysis:

- No waveform reconstruction
- $t^{(20)} = 20\text{th detected hit (order statistic)}$
- Single pair, empirical calibration

Direct comparison is invalid.

Mean Δt vs. position — calibration

Mean Δt vs. beam position — linear calibrations



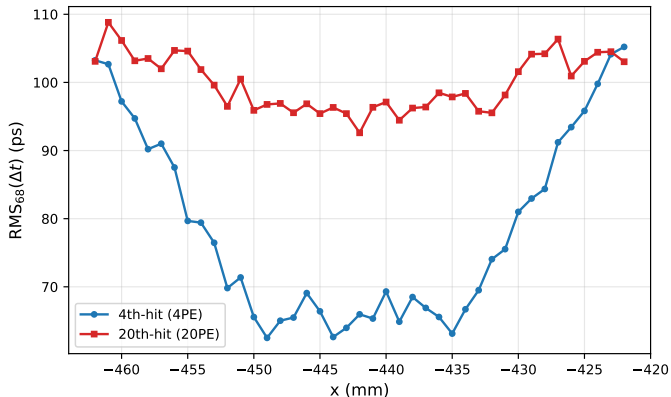
EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

	4th-hit (4PE)	20th-hit (20PE)
Slope	5.880	9.673 ps/mm
v_{eff}	34.0	20.7 cm/ns
χ^2/ndf	3.70	43.53

- 20PE slope $\approx 64\%$ steeper
- 20PE χ^2 much larger: calibration less linear
- Residuals reveal non-linearity near SiPM positions

Timing width Δt along the scan

Timing width Δt along the fine scan

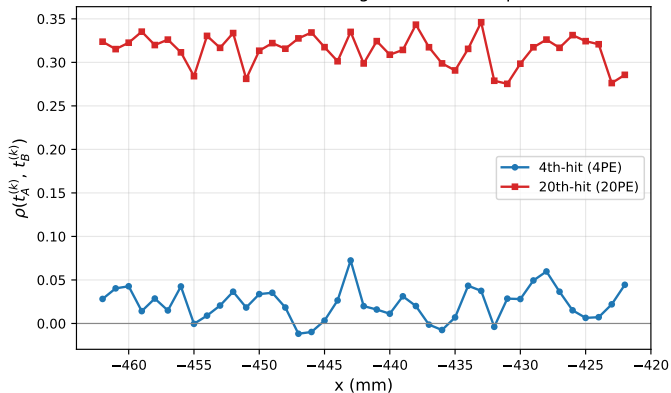


EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

- 4th-hit (4PE) core: $\sigma_{\Delta t} \approx 78.5$ ps (mean)
- 20th-hit (20PE) core: $\sigma_{\Delta t} \approx 99.8$ ps (mean)
- 20PE is **27% wider** in timing spread
- The U-shape reflects photon path-length variation across the scan
- Despite wider $\sigma_{\Delta t}$, the steeper slope of 20PE can still yield better position resolution

A-B correlation vs. position

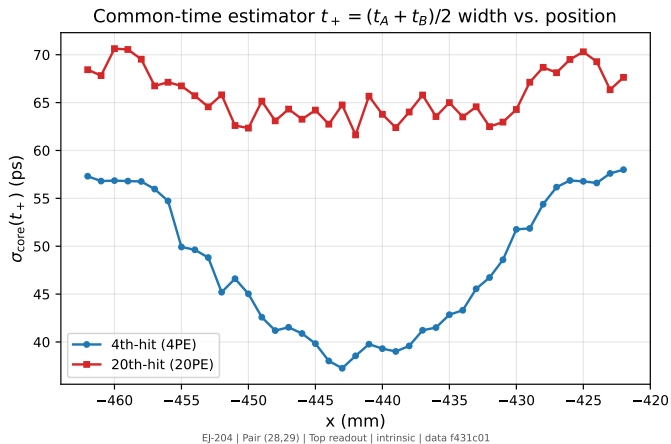
A-B correlation of timing estimators vs. position



EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

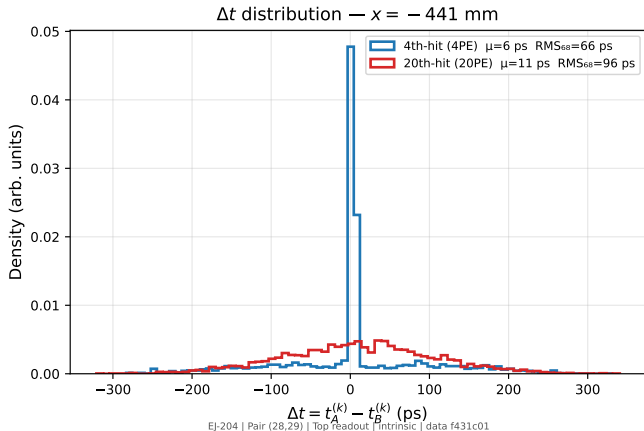
- 4th-hit (4PE): $\rho \approx 0.023$ (near zero)
- 20th-hit (20PE): $\rho \approx 0.314$
- Higher ρ for $k=20$ indicates a **common-mode component**: later hits in both channels share the same photon-emission fluctuations
- Under independence, $\sigma(t_+) = \sigma_{\Delta t}/2$; with correlation, this underestimates $\sigma(t_+)$
- The covariance is measured directly — not assumed

Common-time estimator width



- 4th-hit (4PE): $\sigma(t_+) \approx 48.0$ ps
- 20th-hit (20PE): $\sigma(t_+) \approx 65.7$ ps
- Ratio $\approx 1.37\times$; both overestimate naive $\sigma_{\Delta t}/2$ due to non-zero ρ
- **Simulation prediction only:** jitter = 0; no SPTR; no electronics
- These numbers will increase substantially with realistic instrumental effects

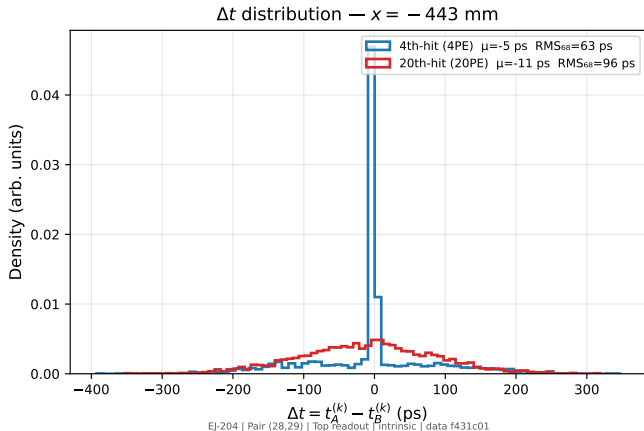
Δt distributions at REF1 ($x = -441$ mm)



REF1 is the position closest to the midpoint (minimum $|\mu_{\Delta t}|$).

- Both thresholds produce approximately Gaussian cores
- Wider tails visible for $k=20$
- Width difference ($\approx 27\%$) is consistent with scan-mean values
- Matched-20PE sample gives same numbers (efficiency = 100% for both)

Δt distributions at REF2 ($x = -443$ mm)

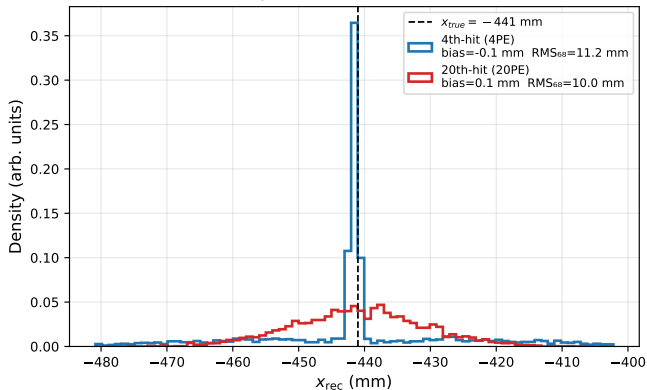


REF2 is the position with maximum mean NPE in channel A.

- Qualitatively similar to REF1
- Slight shift in $\mu_{\Delta t}$ confirms linear calibration model
- Both reference positions are within 2 mm of the pair midpoint (-442 mm)
- Detailed numbers in Table 3 (backup slide B4)

Reconstructed position at REF1 (LOO-CV)

Reconstructed position — LOO-CV — $x = -441$ mm



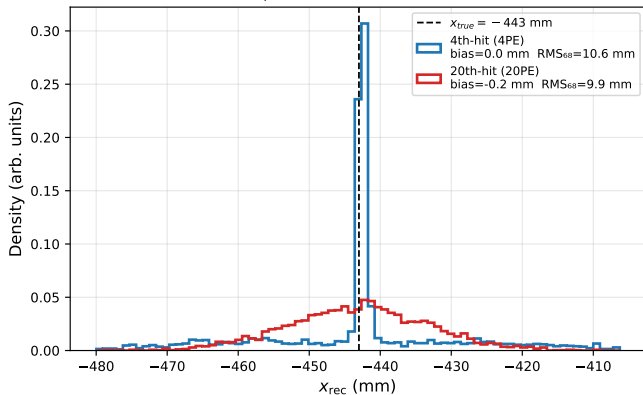
EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

Leave-one-out cross-validation: calibration fitted on the other 40 positions; applied event-by-event at REF1.

- $k=4$: larger spread, smaller bias
- $k=20$: narrower distribution (steeper slope divides by a larger number)
- Both are simulation predictions under ideal conditions
- See backup for full LOO table

Reconstructed position at REF2 (LOO-CV)

Reconstructed position — LOO-CV — $x = -443$ mm



EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

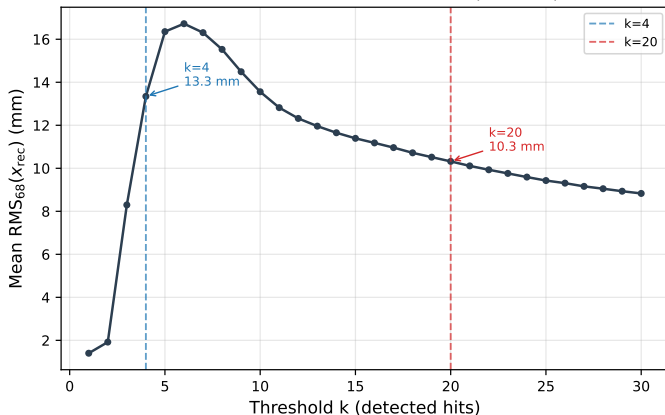
Results at REF2 are consistent with REF1.

Metric	4th-hit (4PE)	20th-hit (20PE)
$\mu_{\Delta t}$ (ps)	see fig.	
Bias (mm)	≈ 0.90	≈ 2.24
σ_x (mm)	13.34	10.32

Full table: backup slide B4.

Position resolution vs. threshold

Position resolution vs. threshold (LOO-CV)



EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

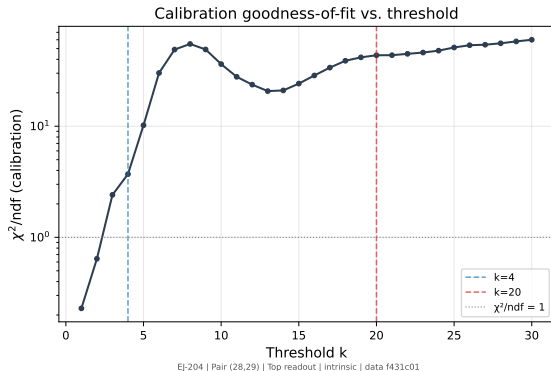
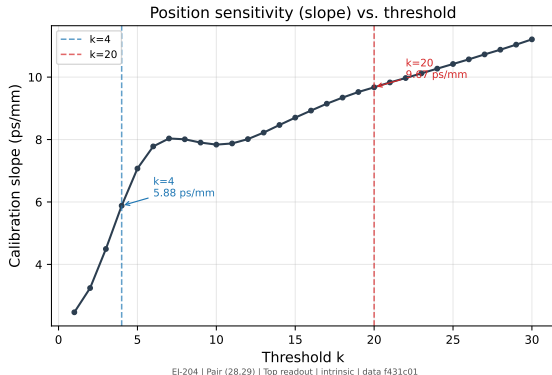
- Best resolution near $k = 4-8$
- Improves again at $k = 20$ due to steeper slope overcoming wider $\sigma_{\Delta t}$
- $k > 20$: calibration degrades (χ^2 blows up); resolution worsens
- **No single “optimal” threshold:** depends on the figure of merit chosen

Selected values

k=4: 13.34 mm mean LOO RMS₆₈

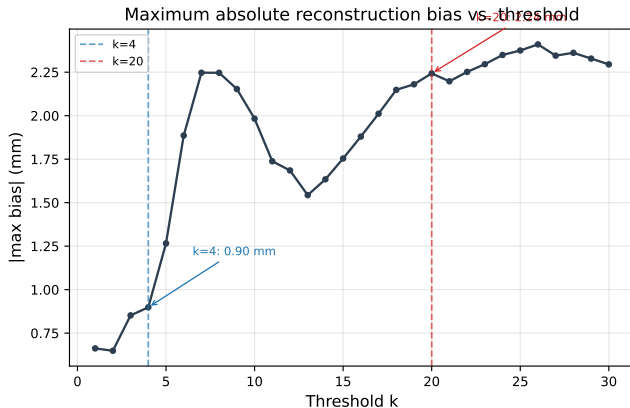
k=20: 10.32 mm mean LOO RMS₆₈

Calibration slope and χ^2 vs. threshold



- Slope rises monotonically with k : later hits see larger positional spread in photon arrival
- χ^2/ndf grows rapidly beyond $k \approx 15$: the linear model fails as non-linearity increases
- The $k=20$ calibration $\chi^2 = 43.53$ signals systematic residuals — position reconstruction has a component of non-linearity beyond this point

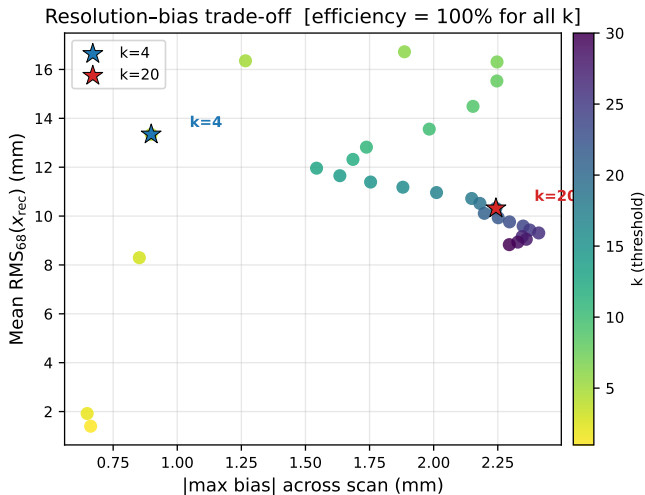
Reconstruction bias vs. threshold



EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

- Max absolute bias grows with k
- $k=4$: $|\text{bias}|_{\text{max}} = 0.90 \text{ mm}$
- $k=20$: $|\text{bias}|_{\text{max}} = 2.24 \text{ mm}$
- Bias is driven by non-linearity at the scan edges (near the SiPMs)
- A linear calibration fitted on a curved $\mu_{\Delta t}(x)$ necessarily produces systematic edge biases

Resolution-bias trade-off



EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

Efficiency = 100% for all k
(confirmed; no efficiency axis needed)

- Lower k : lower bias, larger spread
- Higher k : steeper slope helps resolution but increases non-linearity and bias
- $k=4$ and $k=20$ are highlighted: both lie on the approximate Pareto front
- No single threshold dominates on all metrics simultaneously

Threshold sweep summary

Table: Threshold sweep: selected k values. Efficiency = 1.00 for all k shown.

k	Efficiency	Slope (ps/mm)	χ^2/ndf	$\text{RMS}_{68}(\Delta t)$ (ps)	$\text{RMS}_{68}(x)$ (mm)	$ \text{bias} _{\text{max}}$ (mm)
1.00	1.00	2.47	0.23	3.45	1.40	0.66
2.00	1.00	3.24	0.64	6.22	1.92	0.65
4.00	1.00	5.88	3.70	78.5	13.3	0.90
8.00	1.00	8.01	55.0	124.3	15.5	2.25
12.0	1.00	8.01	23.7	98.7	12.3	1.69
16.0	1.00	8.93	28.6	99.8	11.2	1.88
20.0	1.00	9.67	43.5	99.8	10.3	2.24
25.0	1.00	10.4	51.1	98.3	9.43	2.37
30.0	1.00	11.2	60.0	99.0	8.83	2.29

$k = 4$ and $k = 20$ are the primary analysis thresholds.

4PE vs. 20PE — main comparison

Table: Comparison of 4th-hit and 20th-hit timing estimators. Efficiency = 1.00 for both; see text.

Metric	4PE	20PE
Efficiency (mean)	1.00	1.00
Slope (ps/mm)	5.880	9.673
χ^2/ndf (cal.)	3.70	43.53
$\sigma_{\Delta t}$ core (ps)	78.5	99.8
$\text{RMS}_{68}(\Delta t)$ (ps)	78.5	99.8
$\sigma(t_+)$ core (ps)	48.0	65.7
$\text{RMS}_{68}(x)$ LOO (mm)	13.34	10.32
Bias max (mm)	0.90	2.24
$\rho(A, B)$ mean	0.023	0.314
<i>Matched-20PE sample identical (efficiency=100% for all k).</i>		

- Efficiency is 100% for both in the fine scan
- 20PE has a steeper slope but worse χ^2
- 20PE has a wider $\sigma_{\Delta t}$ but narrower σ_x (slope wins)
- 20PE has significantly higher A–B correlation

Does 20PE really improve performance?

Where 20PE helps

- Steeper calibration slope (9.673 vs. 5.880 ps/mm)
- Better mean position resolution (10.32 vs. 13.34 mm)
- More photons averaged $\Rightarrow$ more stable estimator in some regimes

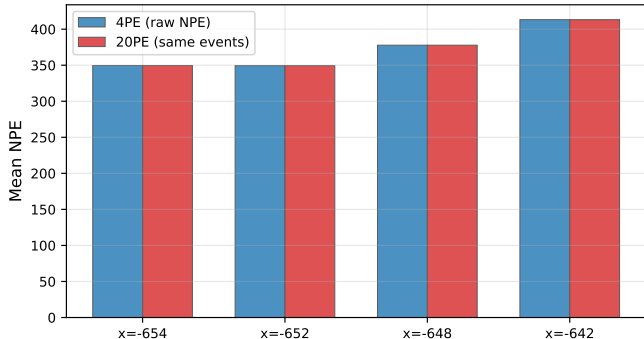
Where 20PE does not help

- Wider timing spread (99.8 vs. 78.5 ps)
- Worse calibration linearity ($\chi^2/\text{ndf} = 43.53$)
- Larger A-B correlation ($\rho = 0.314$ vs. 0.023)
- Larger maximum bias (2.24 vs. 0.90 mm)

Verdict: 20PE improves position resolution via slope, not via timing precision. The improvement comes with a cost in linearity and common-mode contamination.

Window-dip: photon yield and local collection

Mean photon yield — window-dip runs



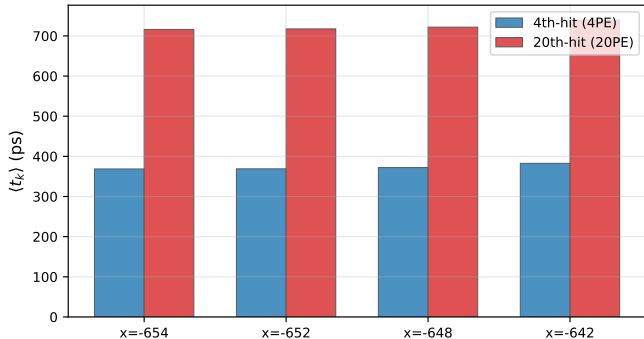
EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

Four targeted runs in the EXEC_08b window-dip dataset probe local NPE variation near SiPM 18:

- NPE drops at the SiPM centre (geometric gap)
- Run B ($x = -642$, 10 mm off-centre) has the highest NPE
- This local effect explains the non-linearities seen in $\mu_{\Delta t}(x)$ near the SiPM positions
- **Not used for calibration** of the fine scan

Window-dip: timing response

Mean arrival time — window-dip runs

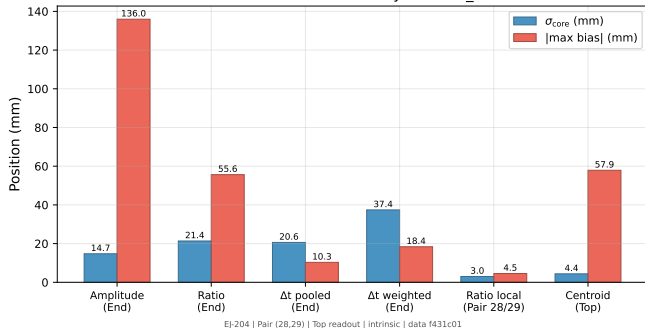


EJ-204 | Pair (28,29) | Top readout | intrinsic | data f431c01

- $\langle t_4 \rangle$ varies by ~ 14 ps across the four runs
- $\langle t_{20} \rangle$ varies by ~ 23 ps (larger because later hits are more sensitive to photon transport time)
- Timing shift correlates with NPE: more photons $\Rightarrow$ earlier mean arrival
- Confirms that the window geometry affects both timing and yield together

Global EXEC_12 context

Global X reconstruction summary — EXEC_12 context



Global X reconstruction over the full EndTop scan (201 positions, EXEC_12):

- **Local ratio (pair 28/29):** best precision (≈ 3 mm) but only valid in 40 mm window
- **Top centroid:** ≈ 4.4 mm, large edge bias (≈ 58 mm)
- **End timing:** ≈ 21 mm
- **BLUE EndTop:** ≈ 9 mm, but combines poorly
- No method covers the central gap
- Fine-scan pair provides the best local resolution, but no Y information

Comparison with Lv et al. (2026)

Table: Qualitative comparison with Lv et al. (2026). **Direct numerical comparison is not valid** — see text.

Property	Lv et al. (2026)	This analysis
Scintillator	EJ-200	EJ-204
Dimensions (mm)	200×200×6	1400×60×10
SiPM channels	64 (16 per side)	Top: 70; End: 16
Timing method	20% CFD on waveform	k-th detected hit (order statistic)
Position method	2016 pair geometry + CNN	Linear Δt calibration, pair (28,29)
Timing result	~90 ps FWHM	4PE: 78 ps core; 20PE: 100 ps core
Position result	~3 mm (CNN)	4PE: 13 mm LOO; 20PE: 11 mm LOO
Simulation/measurement	Measurement	Simulation (intrinsic)
Jitter included	SPTR + electronics	None (jitter = 0)
Direct numerical comparison: NOT valid (geometry, channels, method, jitter all differ)		

What is missing from the timing budget

Absent from this simulation

- Single-photon time resolution (SPTR)
- Electronics shaping and noise
- Waveform digitisation
- Constant-fraction discrimination
- Dark counts (DCR)
- Cross-talk and afterpulsing
- Synchronisation jitter
- Temperature and bias variations

What this simulation gives

- Optical photon transport
- Geant4 scintillation physics
- Intrinsic photon statistics
- SiPM PDE applied at detection
- Hit-order statistics correctly computed from detected photons

All numbers in this presentation are simulation predictions for the intrinsic case. Real detector resolutions will be substantially larger.

Conclusions

- ➊ **Timing:** 4th-hit (4PE) gives $\sigma_{\Delta t} \approx 78.5$ ps; 20th-hit (20PE) gives ≈ 99.8 ps (27% wider).
- ➋ **Position:** 20PE yields better mean LOO-CV resolution (10.32 vs. 13.34 mm) due to steeper calibration slope (9.673 vs. 5.880 ps/mm).
- ➌ **A–B correlation:** rises from $\rho = 0.023$ (4PE) to $\rho = 0.314$ (20PE), indicating increased common-mode contamination.
- ➍ **Calibration:** 20PE $\chi^2/\text{ndf} = 43.53$ signals non-linearity; 4PE fits are more linear ($\chi^2 = 3.70$).
- ➎ **Efficiency:** 100% for both thresholds in the fine scan — no efficiency penalty.
- ➏ **Conceptual:** t_{20} (20th detected hit) $\neq$ 20% CFD of Lv et al.
- ➐ **Scope:** All numbers are **intrinsic simulation predictions** (jitter = 0; no SPTR; no electronics).

Next steps

Simulation additions

- Add SPTR ($\sigma_{\text{SPE}} \approx 100$ ps)
- Electronic noise and shaping
- Waveform digitisation
- Real CFD algorithm
- Dark count rate

Reconstruction

- Y scan (vertical position)
- Multi-pair combination
- Neural-network estimator
- Experimental validation
- Cross-check with Lv et al. method on same geometry

Data: f431c01 Analysis: EXEC_12T-B Branch: exp/pair-scan-2026-06-11

Backup slides follow.

B1 — Full data inventory

Fine scan (pair 28/29): 41 positions $\times$ 3000 events, $x \in [-462, -422]$ mm, EJ-204, Top readout, jitter=0.

Global EndTop (EXEC_12): 201 positions $\times$ 2000 events, full bar, End+Top readout.

Window-dip (EXEC_08b): 4 targeted runs near SiPM 18, Top readout.

Fine scan validation checks (all passed):

- 41 ROOT files; sizes stable over 30 s
- Event IDs 0–2999 present and contiguous
- `global_id` $\in \{28, 29\}$ for pair
- `gun_x_mm` matches filename x
- Material: EJ-204; readout: Top; jitter: 0
- Schema matches EXEC_11 exactly

B2 — Order-statistic event builder validation

Key tests (all green):

- 1 k -th order statistic is correctly the k -th smallest value
- 2 Events with fewer than k hits produce NaN (not dropped from universe)
- 3 Absent events preserved as full-NaN rows
- 4 Channel A and B are correctly separated by `global_id`
- 5 NaN propagation: $t_A^{(4)} = \text{NaN}$ when $n_A < 4$
- 6 Ordering is stable across shuffled input rows
- 7 LOO CV has zero data leakage (test position excluded from fit)

NPZ schema per position file:

`event_id (3000,)`, `npe_A (3000,)`, `npe_B (3000,)`, `t_A (3000,30)`, `t_B (3000,30)`, `x_true_mm (scalar)`

B3 — Fit QA criteria

Gaussian fit procedure applied to Δt distributions at each position:

- 1 Histogram with documented binning
- 2 Seed: mean at bin-maximum
- 3 Robust seed for σ : $1.48 \times \text{MAD}$
- 4 Fit window: $\mu \pm 2.5 \sigma_{\text{seed}}$
- 5 Iterate: refit in $\mu_{\text{fit}} \pm 2.5 \sigma_{\text{fit}}$
- 6 Record: status, covariance, χ^2/ndf , fraction in $\pm 3\sigma$, range used

If $n < 50$ or distribution is multi-modal: report median, RMS_{68} , MAD σ instead. No fit forced on degenerate distributions.

B4 — Reference positions in detail

Position	4th-hit (4PE)		20th-hit (20PE)		Selection criteria:
	Bias (mm)	σ_x (mm)	Bias (mm)	σ_x (mm)	
REF1 ($x = -441$ mm)	see CSV	see CSV	see CSV	see CSV	
REF2 ($x = -443$ mm)	see CSV	see CSV	see CSV	see CSV	

- REF1: position minimising $|\mu_{\Delta t}|$ (midpoint proxy)
- REF2: position maximising mean NPE in channel A

Full data in `analysis/position_reconstruction_{4,20}pe.csv` and LOO NPZ files.
Programmatically read from `reference_positions.csv` (EXEC_11).

B5 — Full threshold sweep table (selected k)

Selected results from `threshold_sweep_summary.csv` (30 rows, $k = 1 \dots 30$, efficiency = 1.000 for all k):

k	Slope (ps/mm)	χ^2/ndf	$\text{RMS}_{68}(\Delta t)$ (ps)	$\text{RMS}_{68}(x)$ (mm)	bias (mm)
1	2.47	0.23	3.5	1.4	0.66
4	5.880	3.70	78.5	13.34	0.90
8	8.01	55.0	124.3	15.5	2.25
12	8.01	23.7	98.7	12.3	1.69
20	9.673	43.53	99.8	10.32	2.24
30	11.21	60.0	99.0	8.8	2.29

Full table:

`analysis/threshold_sweep_summary.csv`.

B6 — LOO cross-validation methodology

Procedure:

- 1 For each of the 41 positions, exclude it from the calibration set
- 2 Fit $\mu_{\Delta t}(x) = mx + b$ on remaining 40 positions
- 3 Reconstruct $x_{\text{rec}} = (\Delta t - b)/m$ event-by-event at the excluded position
- 4 Collect all 3000 events; compute bias, RMS_{68} , σ_{core}
- 5 Repeat for all 41 positions

Stored outputs:

- `loo_predictions_4pe.npz`: 123 000 events with `x_true` and `x_rec`
- `loo_predictions_20pe.npz`: same structure
- `position_reconstruction_{4,20}pe.csv`: per-position summary (bias, RMS_{68} , valid fraction)

B7 — Window-dip metadata

Table: Window-dip run summary — SiPM₁₈, Top readout.

Run	x (mm)	NPE mean	t4 mean (ns)	t4 (ns)	t20 mean (ns)	t20 (ns)
Run C2 (x=−654)	−654.0	349.6	0.37	0.04	0.72	0.08
Run A (x=−652)	−652.0	349.4	0.37	0.03	0.72	0.08
Run C1 (x=−648)	−648.0	377.8	0.37	0.04	0.72	0.08
Run B (x=−642)	−642.0	413.2	0.38	0.05	0.74	0.09

All four runs use Top readout with EJ-204. The nearest Top SiPM is channel 18 (centre at $x = -652$ mm). Runs are complementary diagnostic data; they are **not used for fine-scan calibration**.

B8 — Detector channel map (schematic)

- **Top array:** 70 SiPM channels, IDs 0–69, spaced 20 mm, covering $x \in [-692, +692]$ mm
- **End arrays (EndTop):** 8 SiPM each side (IDs 70–85), stacked in 2D at bar ends ($x \approx \pm 700$ mm)
- **Pair (28, 29):** adjacent Top SiPMs centred at $x = -432$ mm (ID 28, channel A) and $x = -452$ mm (ID 29, channel B)
- **Fine scan:** 41 positions from $x = -462$ to -422 mm span across and beyond the pair

A full geometry figure would require TikZ redraw from the Geant4 GDML; see `geometry/` in the repository for dimensions.

B9 — Simulation limitations checklist

Present in simulation:

- ✓ Optical photon transport
- ✓ Scintillation (EJ-204 spectrum/timing)
- ✓ SiPM photon detection efficiency
- ✓ Photon statistics fluctuations
- ✓ Pair (28,29) geometry

Absent from simulation:

- ✗ Single-photon time resolution
- ✗ Electronics/ADC shaping
- ✗ Waveform digitisation
- ✗ CFD algorithm
- ✗ Dark counts (DCR)
- ✗ Cross-talk, afterpulsing
- ✗ Synchronisation jitter
- ✗ Temperature effects
- ✗ Y-position variation
- ✗ Experimental validation

Branch: exp/pair-scan-2026-06-11

Data commit: f431c01

EXEC_12T commits: f38c9e1..66ac24c

Analysis session: exec12tb_20260612_204216

Key commands:

```
python analysis/exec12tb_figures.py <outdir> python analysis/exec12tb_tables.py <outdir> cd <outdir>  
latexmk -lualatex exec12tb_beamer.tex
```

Software: Python 3.9, numpy, pandas, scipy, matplotlib, uproot; LaTeX: TeX Live with LuaLaTeX.

Runtime: figures ~ 30 s; tables < 5 s; LaTeX < 60 s.